

8.2 Review Activity – Rational Function Analysis

Advanced Algebra 2

I can identify asymptotes, holes, and intercepts of rational functions and match graphs to equations.

Name: _____

Period: _____

TEKS: 2A.2A, 2A.6G, 2A.6K

Review for 8.2 Test

Part 1: Name the Asymptotes

Find the VA, HA, and x-intercept(s) for each function.

1. $f(x) = \frac{3}{x-5}$

VA: _____ HA: _____ x-int: _____

4. $f(x) = \frac{5x-10}{2x+8}$

VA: _____ HA: _____ x-int: _____

2. $f(x) = \frac{2x+6}{x+1}$

VA: _____ HA: _____ x-int: _____

5. $f(x) = \frac{-4}{x^2-9}$

VA: _____ HA: _____ x-int: _____

3. $f(x) = \frac{x^2-4}{x+3}$

VA: _____ HA: _____ x-int: _____

6. $f(x) = \frac{3x^2+6x}{x^2-1}$

VA: _____ HA: _____ x-int: _____

Part 2: Hole or Asymptote?

Factor completely. Determine whether each excluded value creates a **hole** or a **vertical asymptote**.

1. $f(x) = \frac{x^2-4}{x-2}$

Factored: _____ Hole or VA? _____ at $x =$ _____ Simplified: _____

2. $f(x) = \frac{x^2+5x+6}{x^2+x-2}$

Factored: _____ Hole at $x =$ _____ VA: _____ HA: _____

3. $f(x) = \frac{x^2-x-6}{x^2-9}$

Factored: _____ Hole at $x =$ _____ VA: _____ HA: _____

Key Question: What is the difference between a hole and a vertical asymptote? How can you tell from the factored form?

Part 3: Horizontal Asymptote Blitz

Find the HA for each. Use “Who Wins?” Watch for **fractional answers**!

4. $f(x) = \frac{7}{x^2+1}$

HA: _____

8. $f(x) = \frac{-2x+7}{6x-1}$

HA: _____

5. $f(x) = \frac{4x-3}{x+5}$

HA: _____

9. $f(x) = \frac{x^2-4x}{2x^2+3x}$

HA: _____

6. $f(x) = \frac{x^3+2}{x^2-1}$

HA: _____

10. $f(x) = \frac{9x}{3x+4}$

HA: _____

7. $f(x) = \frac{3x^2+1}{5x^2-2}$

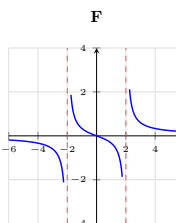
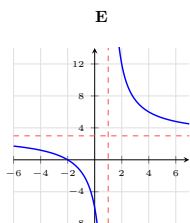
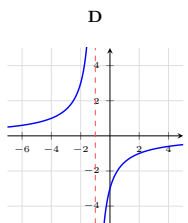
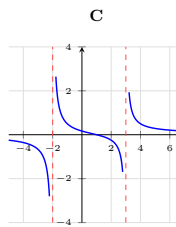
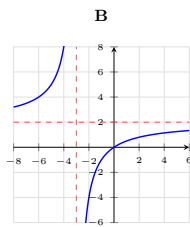
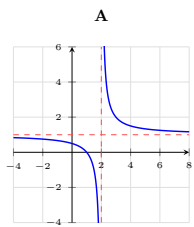
HA: _____

11. $f(x) = \frac{x+1}{x^3-8}$

HA: _____

Part 4: Match the Graph to the Equation

Match each graph (A–F) to its equation. Use the asymptotes and intercepts to decide!



Equations: Write the graph letter (A–F) next to each equation.

___ i. $f(x) = \frac{-3}{x+1}$

___ iii. $f(x) = \frac{1}{x-2} + 1$

___ v. $f(x) = \frac{x-1}{x^2-x-6}$

___ ii. $f(x) = \frac{x}{x^2-4}$

___ iv. $f(x) = \frac{2x}{x+3}$

___ vi. $f(x) = \frac{3x+6}{x-1}$

Part 5: Complete Analysis

Find **all** key features. Factor first!

4. $f(x) = \frac{2x^2 - 8}{x^2 - x - 2}$

Factored: _____

VA:	HA:	Hole:	x-Intercept(s):

Domain: _____

5. $f(x) = \frac{x^2 + 3x - 10}{x^2 - 25}$

Factored: _____

VA:	HA:	Hole:	x-Intercept(s):

Domain: _____

8.2 Review Activity

ANSWER KEY

Part 1: Name the Asymptotes

- $\frac{3}{x-5}$: **VA:** $x = 5$ **HA:** $y = 0$ (bottom wins) **x-int:** None (constant numerator)
- $\frac{2x+6}{x+1}$: **VA:** $x = -1$ **HA:** $y = 2$ (tie: $2/1$) **x-int:** $(-3, 0)$ [set $2x + 6 = 0$]
- $\frac{x^2-4}{x+3}$: Factor num: $(x-2)(x+2)$. **VA:** $x = -3$ **HA:** No HA (top wins) **x-int:** $(2, 0)$ and $(-2, 0)$
- $\frac{5x-10}{2x+8}$: **VA:** $x = -4$ **HA:** $y = \frac{5}{2}$ (tie: $5/2$) **x-int:** $(2, 0)$
- $\frac{-4}{x^2-9}$: Factor: $\frac{-4}{(x-3)(x+3)}$. **VA:** $x = 3$ and $x = -3$ **HA:** $y = 0$ **x-int:** None
- $\frac{3x^2+6x}{x^2-1}$: Factor: $\frac{3x(x+2)}{(x-1)(x+1)}$. **VA:** $x = 1$ and $x = -1$ **HA:** $y = 3$ (tie: $3/1$) **x-int:** $(0, 0)$ and $(-2, 0)$

Part 2: Hole or Asymptote?

- $\frac{(x+2)(x-2)}{x-2}$: $(x-2)$ cancels $\rightarrow$ **Hole at** $x = 2$. Simplified: $x + 2$
- $\frac{(x+2)(x+3)}{(x+2)(x-1)}$: $(x+2)$ cancels $\rightarrow$ **Hole at** $x = -2$. $(x-1)$ stays $\rightarrow$ **VA at** $x = 1$. **HA:** $y = 1$ (tie: $1/1$)
- $\frac{(x-3)(x+2)}{(x-3)(x+3)}$: $(x-3)$ cancels $\rightarrow$ **Hole at** $x = 3$. $(x+3)$ stays $\rightarrow$ **VA at** $x = -3$. **HA:** $y = 1$

Key Question: A hole occurs when a factor *cancels* (appears in both numerator and denominator). A VA occurs when a factor *stays* in the denominator after canceling. At a VA, the graph goes to $\pm\infty$; at a hole, there's just a missing point.

Part 3: HA Blitz

- | | |
|---|--------------------------------------|
| 10. Bottom wins $\rightarrow y = 0$ | 14. Tie: $-2/6 \rightarrow y = -1/3$ |
| 11. Tie: $4/1 \rightarrow y = 4$ | 15. Tie: $1/2 \rightarrow y = 1/2$ |
| 12. Top wins $\rightarrow$ No HA | 16. Tie: $9/3 \rightarrow y = 3$ |
| 13. Tie: $3/5 \rightarrow y = 3/5$ | 17. Bottom wins $\rightarrow y = 0$ |

Part 4: Graph Matching

- | | | |
|--|---|---|
| i. $\frac{-3}{x+1} \rightarrow \mathbf{D}$ | iii. $\frac{1}{x-2} + 1 \rightarrow \mathbf{A}$ | v. $\frac{x-1}{x^2-x-6} \rightarrow \mathbf{C}$ |
| ii. $\frac{x}{x^2-4} \rightarrow \mathbf{F}$ | iv. $\frac{2x}{x+3} \rightarrow \mathbf{B}$ | vi. $\frac{3x+6}{x-1} \rightarrow \mathbf{E}$ |

Part 5: Complete Analysis

- $\frac{2x^2-8}{x^2-x-2} = \frac{2(x-2)(x+2)}{(x-2)(x+1)}$. Cancel $(x-2) \rightarrow$ Simplified: $\frac{2(x+2)}{x+1}$
VA: $x = -1$ **HA:** $y = 2$ **Hole:** $x = 2$ **x-int:** $(-2, 0)$
 Domain: $(-\infty, -1) \cup (-1, 2) \cup (2, \infty)$
- $\frac{x^2+3x-10}{x^2-25} = \frac{(x+5)(x-2)}{(x+5)(x-5)}$. Cancel $(x+5) \rightarrow$ Simplified: $\frac{x-2}{x-5}$
VA: $x = 5$ **HA:** $y = 1$ **Hole:** $x = -5$ **x-int:** $(2, 0)$
 Domain: $(-\infty, -5) \cup (-5, 5) \cup (5, \infty)$